RITHWIK RACHARLA

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OBJECTIVE

Seeking a Software Engineering internship to apply strong foundations in data structures, algorithms, backend development, and full-stack engineering to build reliable software systems.

EDUCATION

B.Tech in Computer Science and Engineering – VNR VJIET Expected 2028
CGPA: **8.93/10**

SKILLS

Programming Languages	Java, Python, C++, JavaScript
Core Concepts	Data Structures & Algorithms, OOP, OS, DBMS, Computer Networks
Development	React, Node.js, Express.js, REST APIs, JWT, MongoDB, MySQL, SQL
Tools & Deployment	Git, GitHub, VS Code, Vercel, Render, Sentry, PostHog

PROJECTS

MedSafe – Medication Safety Checker GitHub *Python, FastAPI, Groq, Pytest*

- Built medication-safety research prototype: parses prescription text, normalizes drug names, and flags drug–drug interactions, dose issues, and allergy conflicts via rule engine.
- Designed conservative pipeline where evidence-backed rule results stay authoritative; optional Groq explanations are non-authoritative, so LLM output cannot override validated checks.
- Sourced interaction and labeling data from DDInter and DailyMed with alias normalization; Pytest suite covers parsing, rule validation, and explanation behavior.

Distill – AI News Digest GitHub – Live *Node.js, transformers.js, Groq, Telegram/WhatsApp APIs, Vercel*

- Built personalized daily AI-news digest delivered on Telegram (LIVE) and WhatsApp (in testing): RSS fetch, interest-based filtering, Groq LLM summaries with "why read this" notes.
- Designed dedup pipeline using MiniLM embedding clustering instead of exact-match, merging same-event stories across outlets; confidence tiers flag single-source claims.
- Added multi-user support via Clerk Google OAuth and one-time-code Telegram linking. Deployed on Vercel with Sentry and PostHog monitoring.

CureConnect – Offline-First Healthcare PWA GitHub – Live *React, TypeScript, Node.js, Express, MongoDB, Firebase Auth*

- Built installable offline-first PWA for traveling doctors to manage patient records, with Express + MongoDB backend and Firebase authentication.
- Implemented IndexedDB persistence with background sync: CRUD works offline and queued updates sync on reconnect.
- Added client-side AES-GCM encryption (Web Crypto API) for sensitive medical fields before storage.

COMPETITIVE PROGRAMMING

- Solved 500+ DSA problems across LeetCode (200+), Codeforces, CodeChef, SmartInterviews (200+).
- Contest maximum ratings: LeetCode **1672**, CodeChef **1414 (2⋆)**.

CERTIFICATIONS & TRAINING

- HIVE – SmartInterviews, Data Structures & Algorithms.
- CSI BuildWithRAG Workshop – Built a RAG application on sports data.
- **Industrial Training** – MERN Stack (12 months), VNR VJIET; Agentic AI (2 months), VNR VJIET.